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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIO

GCE O Level

MARK SCHEME for the November 2005 question

4024 MATHEMATICS

4024/01 Paper 1 maximum raw mark 80

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All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

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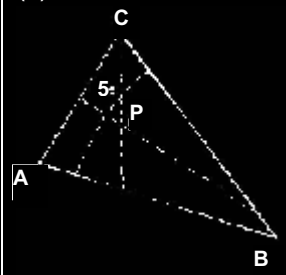
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1	**(a)**	2.44	**1**	
	(b)	(0).021	**1**	
2	**(a)**	$\frac{9}{20}$	**1**	
	(b)	$\frac{2}{15}$ c.a.o.	**1**	
3	**(a)**	$\frac{3}{8}$ or $\frac{6}{16}$ only	**1**	
	(b)	30	**1**	
4	**(a)**	M, S, L	**1**	
	(b)	20	**1**	
5	**(a)**	$\frac{1}{4}$ c.a.o.	**1**	
	(b)	2.4×10^6 c.a.o.	**1**	
6	**(a)**	190	**1**	
	(b)	$\frac{1}{2}(n+1)(n+2)$ o.e. (seen)	**1***	Accept $(n+1+1)$
			[12]	
7		$\frac{90000}{50x60}$ M1		
		30 A1	**2***	
8	**(a)**	73	**1**	
	(b)	31 f.t. their 73 − 42	**f.t. 1**	
	(c)	318	**1**	
9	**(a)**	Fig. 6	**1**	
	(b)	Fig. 4	**1**	
	(c)	Fig. 2	**1**	
10	**(a)**	75	**1**	
	(b)	$\frac{360}{180-165}$ or $(2n-4)\,90 = 165n$ M1	**1**	o.e.
		24 A1	**2***	
			[11]	
11	**(a)**	$5x(x-2)$	**1**	
	(b)	4	**1**	
	(c)	0 or −2	**1**	
12	**(a)**	$A\hat{C}B = C\hat{D}A$ *and* $B\hat{A}C = A\hat{C}D$	**1**	Any irrelevant or wrong information = 0
		$\Rightarrow \Delta$s similar	**1**	
	(b)	$\frac{7}{AD} = \frac{4}{6}$ *or* $\frac{6}{9}$ M1		
		10½ A1	**2***	

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13	**(a)**		**1**	
	(b)	**(i)** Squares	**1**	
		(ii) 	**1**	A (
14	**(a)**	$y \geq \frac{1}{2}x$ o.e.	**1**	
	(b)	$-4\frac{1}{2} \leq x < -2$ M1 −4 and −3 A1	**2***	Accept as separate statements
			[12]	
15	**(a)**	$\begin{pmatrix} 0 & 1 \\ -1 & 2 \\ 0 & -3 \end{pmatrix}$	**2**	**SC1** for 4 or 5 elements correct
	(b)	$(1 \quad -1)$	**2**	**SC1** for a (1 x 2) matrix
16	**(a)**	−17	**1**	
	(b)	5	**1**	
	(c)	$\frac{1}{3}(x+5)$	**1**	Allow y etc.
	(d)	3 f.t.	**f.t. 1**	
17	**(a)**	Idea of 100 ± 2.5 *or* 75 ± 2.5 M1		i.e. any one of 97.5, 102.5, 72.5 or 77.5 seen
		340 A1	**2***	
	(b)	$\frac{22.5 \text{ or } 21.5}{2.5 \text{ or } 3.5}$ M1		
		9 A1	**2***	
18	**(a)**	$x = 0$	**1**	
		$y = -2$	**1**	
	(b)	**(i)** 13200	**1**	
		(ii) 500	**1**	
			[16]	
19	**(a)**	219 → 221 incl.	**1**	
	(b)	13	**1**	
	(c)	All 8 points plotted correctly P1 Smooth curve C1	**2**	
	(d)	A – any comparison using curves	**1**	
20	**(a)**	13 − 14	**1**	
	(b)	$\frac{2}{3}$ or 0.66 − 0.67	**1**	
	(c)	**(i)** 500	**1**	
		(ii) 700 f.t. their 500 + 200	**f.t. 1**	
	(d)	 straight line L1 curve C1	**2**	A B from (30,300) to (40, their 500 f.t.) from (40, their 500 f.t.) to (60, their 700)
			[11]	

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21	**(a)**	(4, 4)	**1**	
	(b)	(2½, 2)	**1**	
	(c)	$y = 4$	**1**	
	(d)	$y = ½x - ½$ B1 + B1	**2***	M st
	(e)	20	**1**	
22	**(a)**	(6, 2)	**1**	
	(b)	**(i)** (− 2, 0)	**1**	
		(ii) 90° AC	**1**	
	(c)	(0, −2), (−4, −2) (−6, −6)	**2**	**SC1** for 2 points plotted correctly or 3 points stated
	(d)	$\begin{pmatrix} -\frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$	**1**	
			[12]	
23	**(a)**	(i) 1:2 000 000	**1**	
		(ii) 235 − 237	**1**	
	(b)	 Constructions I L bisect C1 II I bisect M1 III arc B1 The possible positions clearly indicated P1	**4**	I within 2° II within 2° 2 mm III within 2 mm
			[6]	